

Jora Hoist Raise

03/10/2019

Jora Hoist Applicability I

- *Method is applicable for vertical or incline raise.
- *Before starting Jora Raise, both upper level and lower level should be available.
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Preparation I

- *One large hole has to be drilled at centre of proposed raise.
- *Hole diameter should be large enough for rope and rope capel to pass through.
- *While drilling hole, care has to be taken to avoid deviation.
- *Suitable hoist has to be installed at upper level and rope to pass through hole with help of pulley.
- *A cage is suspended at lower level from rope, hanged from upper level.

Preparation II

- *Few round of blasting is done from lower level to initiate raise.
- *When height of raise becomes sufficient for hanging cage, raising is done with help of cage.

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Raising Process I

- *A cage is suspended at lower level from rope, hanged from upper level.
- *Cage has flat roof with fencing to work as working platform for caring out drilling and blasting.
- *From upper level cage is suspended using a steel rope, that can hoist up & down using winze.

Raising Process II

- *Cage is fixed against raise side with help of jack mounted at cage side.
- *For incline raise cage is guided with help of channel rail.
- *This channel rail is mounted at back of raise with help of expansion rock bolt.

Drilling, Blasting I

- *Centre hole serves as free (relief) hole of drilling round.
- *Burn cut (parallel holes) is followed for face blasting.
- *Blasting is done from lower level.
- *Pneumatic Jack hammer suitable for drilling uppers (stopper) is used for drilling upper holes.
- *For drilling, compressed air is provided from lower level hose reel.

Drilling, Blasting II

- *Before blasting, cage is lowered down at lower level & stowed at specific place, so that cage does not get damage by blasting.
- *Hoist rope is taken up to upper level, so that rope does not get damage by blasting.

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Dressing & Mucking I

- *Raise face and side wall are dressed from cage suspended from hoist.
- *For side wall dressing, hook / clamps are fixed at side wall.
- *Broken rocks are removed from bottom of raise with help of loader and mine car.

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Jora Hoist Survey I

- *Proper survey has to be done before drilling centre hole for Jora Lift.
- *Once, raising has been initiated, there is no need to survey.
- *In Jora lift, there is no possibility of changing raise direction.

Jora Hoist Ventilation I

- *Partially, centre hole drilled for Jora lift serves as ventilation return.
- *Compressed air also serve as ventilation air at face.

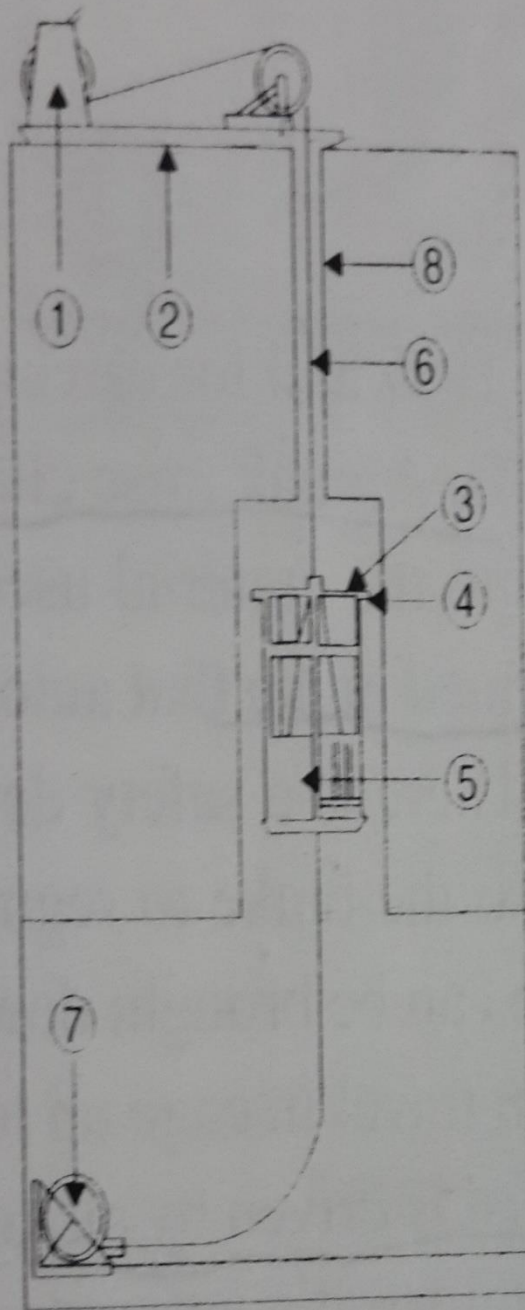
Advantages I

- *There is no possibility for raise to deviate and go into wrong direction.
- *Arrangement is simple, requires least maintenance.
- *Low cost, simple technique.
- *Volume of development work before raising is minimal.
- *Raising process is not effaced by hardness of rock.
- *For weak rock mass, raise can be provided with required support.

Disadvantages I

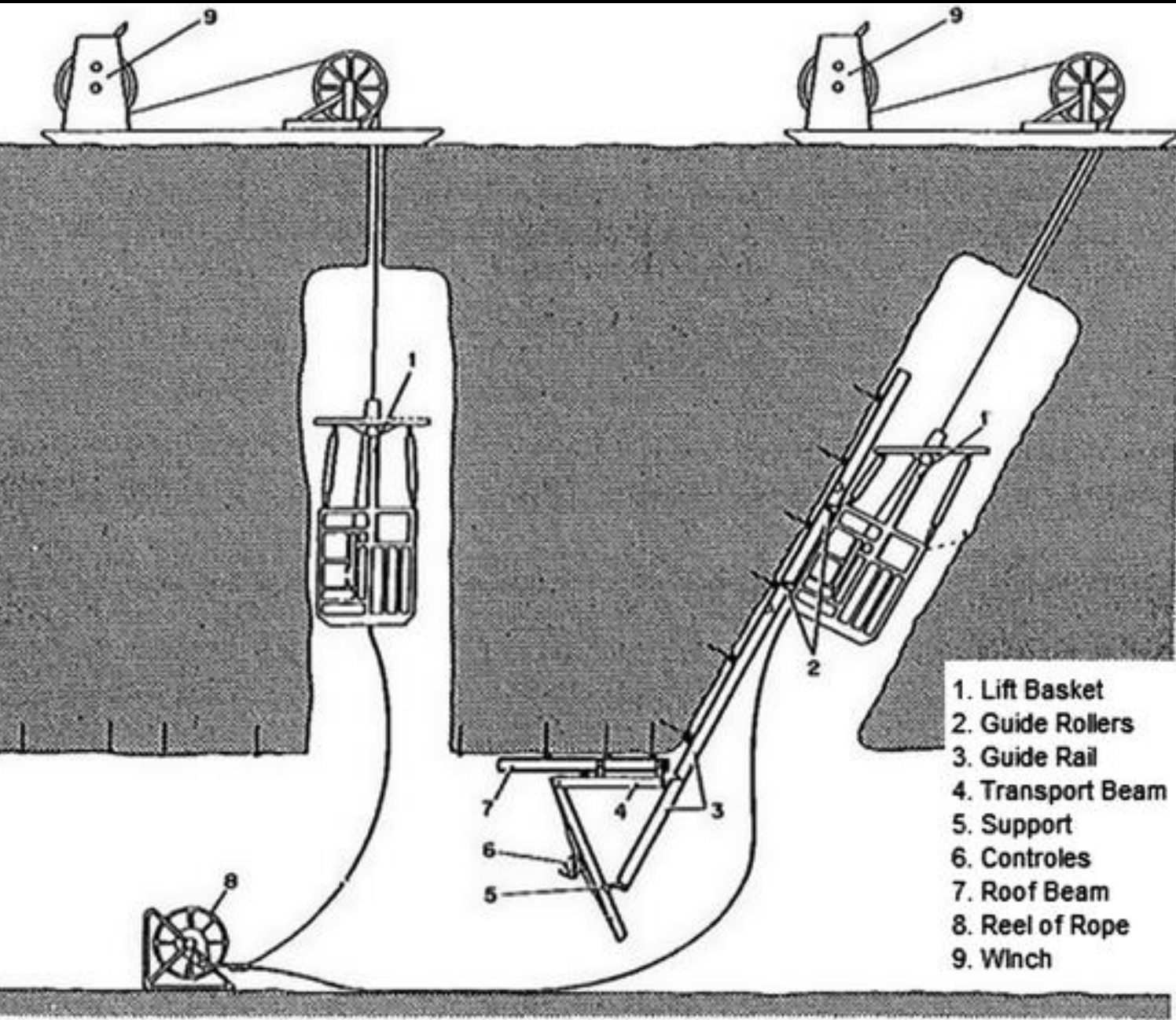
- *Requirement of access at upper level and lower level, both end of intended raise.
- *A large drill machine is required to drill central hole at raise location.
- *Lack of communication in between hoist operator and cage occupant.
- *Damage to hoisting rope.
- *Tedious & risky hoisting operation.

Jora Hoist I



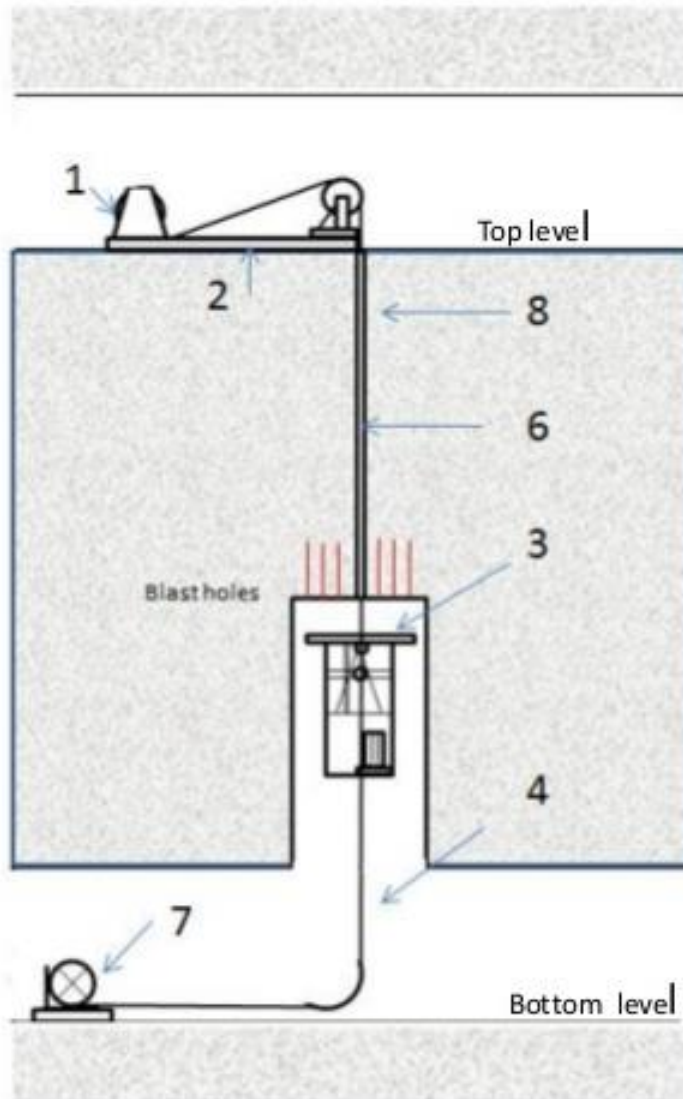
1. Winch for rope
2. Winch skid
3. Drilling platform
4. Telescopic stabilizer
5. Hoist with rope
6. 19 mm dia rope
7. Automatic hole reamer
8. Rope hole

Jora Hoist I



Jora Hoist I

*Rock



- 1- Winch for rope; 2- winch skid; 3- drilling platform; 4- hoist rope;
5- Jora cabin; 6- steel rope; 7- Hole reel; 8- Drill hole for steel rope

3.4.2.2 Raising by Large Diameter Blast Holes

Reference

*Surface and Underground Excavations- Methods, Technique and Equipment. Page no 334- 335